

Estimating Urban Air Pollution Reduction Through Increased Use of Public Transport: A Data Science Approach

Methodology

This project follows a structured and systematic data science methodology to estimate the potential reduction in urban air pollution through increased use of public transport in Delhi.

Data Collection

Air pollution data will be collected from official sources such as the Central Pollution Control Board (CPCB) monitoring stations in Delhi. The dataset will include daily measurements of major air pollutants such as PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, and other available parameters along with date and monitoring station details. To ensure sufficient data volume and analytical depth, observations from multiple monitoring stations and extended time periods will be combined, resulting in more than 1000 records and approximately 20 relevant attributes.

Transport-related factors will be incorporated using publicly available vehicle statistics and structured percentage-based public transport adoption scenarios (e.g., 10%, 20%, and 30% shift from private to public transport).

Input Variables

- Pollutant concentrations (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, etc.)
- Date and monitoring station information
- Derived time-based attributes (month/season)
- Estimated public transport adoption percentage

Output Variable

The output variable will be the estimated pollution level (PM_{2.5}/AQI) under different public transport usage scenarios.

Tools and Techniques

Python programming language will be used along with Pandas and NumPy for data preprocessing, Matplotlib and Seaborn for visualization, and Scikit-learn for regression-based predictive modelling. A “what-if” simulation model will be developed to estimate pollution reduction under varying levels of public transport adoption.

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